

KCPC-CS111

Supplementary Reading

Week 2: Techniques for Investigation

King's Competitive Programming Club

This document contains recommended reading for Workshop 2: Techniques for Investigation. These texts deepen your mathematical problem solving toolkit, spanning heuristics, induction, spatial visualisation, invariance, symmetry, the extreme principle, and bitwise competitive programming.

I. Core Problem-Solving Heuristics

The Art and Craft of Problem Solving (Paul Zeitz)

John Wiley & Sons, 2nd Edition, 2006.

- **Chapter 2: Strategies for Investigation** Psychological strategies, wishful thinking, “make it easier”, drawing diagrams, and identifying patterns from small examples.
- **Chapter 3: Tactics for Solving Problems** Symmetry (geometric and algebraic), parity, and the extreme principle.
- **Chapter 7: Modular Arithmetic & Functional Equations** Techniques for substituting zero or special values into multi-variable equations.

How to Solve It: A New Aspect of Mathematical Method (George Pólya)

Princeton University Press, 1945 / 2004.

- **Part I: In the Classroom** Understanding the problem, devising a plan, carrying out the plan, and looking back.
 - **Heuristic Dictionary** Analogy, auxiliary problems, decomposing and recombining, inductive vs. deductive discovery, and specialization.
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II. The Extreme Principle and Invariants

Problem-Solving Strategies (Arthur Engel)

Springer, 1998.

- **Chapter 2: Coloring Proofs** Grid tilings, dominoes, trominoes, and invariant checkerboard colorings.
 - **Chapter 3: The Extremal Principle** Choosing minimal and maximal elements in finite graphs, discrete geometries, and number systems to produce immediate contradictions.
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III. Bitwise Algorithms and Fibonacci Bases

CP-Algorithms: Bit Manipulation and Bitmasking

<https://cp-algorithms.com/algebra/all-submasks.html>

- Bitwise operations ($\&$, $|$, $\oplus$), population count (`__builtin_popcountll`), and iterating over submasks in $O(3^N)$.
- Structural explanation of $f(2n) = f(n)$ and $f(2n + 1) = f(n) + 1$ as bit-shift operations.

Competitive Programmer's Handbook (Antti Laaksonen)

<https://cses.fi/book/book.pdf>

- **Chapter 10: Bit Manipulation** Bit representations, bitwise operations, and dynamic programming with bitmasks.
- **Chapter 21: Number Theory** Fibonacci properties, Zeckendorf representation, and fast matrix exponentiation.